

LAUWERSOOG, Netherlands

WMO: 062770

Lat: 53.4117N

Lon: 6.1992E

Elev: 10

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.2	25.0	16.3	12.5	24.2	19.9	14.9	25.9	40.5	44.7	37.1	43.8	11.8	100	0.717

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	79.4	67.7	75.2	66.3	72.3	64.8	69.3	76.7	67.4	73.5	65.8	71.0	11.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	97.7	72.8	65.0	92.5	70.8	63.6	88.1	69.2	33.4	77.6	31.9	73.6	30.6	71.2	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.9	30.0	27.0	DB	20.0	88.5	4.5	3.2	16.8	90.8	14.1	92.7	11.6	94.5	8.3	96.9	(4)
(5)			WB	18.9	73.9	4.6	1.7	15.6	75.1	13.0	76.1	10.4	77.1	7.1	78.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	37.5	38.6	42.6	49.2	55.4	60.4	64.3	64.5	60.0	52.4	45.1	39.5		
	DBStd	11.03	6.40	5.98	5.34	5.51	5.20	4.56	4.86	4.20	4.14	5.08	5.18	6.51		
	HDD50	1543	388	320	232	78	8	0	0	0	0	34	158	325		
	HDD65	5281	853	740	693	474	302	153	73	59	159	389	597	789		
	CDD50	1860	0	0	4	55	176	311	443	449	300	110	11	1		
	CDD65	123	0	0	0	1	4	15	51	43	9	0	0	0		
	CDH74	469	0	0	0	5	19	78	186	150	31	0	0	0		
	CDH80	101	0	0	0	0	2	19	46	30	4	0	0	0		
(14) Wind	WSAvg	14.0	15.4	15.0	14.9	13.8	14.4	13.6	13.1	12.1	12.6	13.9	13.7	14.8		
(15) Precipitation	PrecAvg	33.8	2.8	2.2	1.9	1.7	2.2	2.8	3.3	3.7	3.4	3.3	3.3	3.2		
	PrecMax	41.3	5.4	5.5	4.1	4.5	4.2	4.8	7.3	8.0	8.1	6.7	6.1	7.5		
	PrecMin	23.4	0.2	0.8	0.3	0.0	0.5	1.1	1.3	0.7	0.5	1.2	0.4	1.1		
	PrecStd	4.9	1.3	1.0	1.1	0.9	1.0	1.2	1.5	1.9	1.6	1.3	1.3	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	53.3	60.1	72.3	76.5	82.7	85.5	84.3	78.5	68.8	58.7	54.0		
		MCWB	49.7	48.6	51.5	57.7	61.8	69.2	69.7	67.9	66.9	62.1	55.7	51.9		
	2%	DB	49.3	50.0	54.3	65.0	70.5	74.7	79.0	78.0	72.5	63.7	56.0	51.6		
		MCWB	47.5	47.2	49.1	55.6	60.9	65.8	68.8	67.2	64.6	59.4	53.7	49.8		
	5%	DB	47.3	47.8	51.3	60.8	66.6	70.1	74.7	74.0	68.7	60.9	53.6	49.7		
		MCWB	45.6	45.6	48.0	53.8	59.7	63.5	67.0	66.0	63.1	57.6	51.3	48.1		
	10%	DB	45.7	46.0	49.4	57.3	63.3	67.4	71.9	71.1	66.2	59.0	51.8	47.5		
		MCWB	44.0	44.1	46.5	52.2	57.6	62.0	65.4	64.5	61.7	56.3	49.6	45.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.9	49.9	53.7	59.9	65.3	70.4	72.5	71.7	68.4	62.7	56.5	52.4		
		MCDB	51.5	51.9	57.9	69.0	71.2	79.5	81.6	78.8	76.4	67.8	58.3	53.4		
	2%	WB	47.7	47.6	50.4	56.7	62.5	66.8	69.6	68.9	65.7	60.0	54.0	50.1		
		MCDB	48.8	49.4	53.1	62.6	68.5	72.8	77.2	75.5	70.8	62.9	55.7	51.3		
	5%	WB	45.9	45.8	48.5	54.6	60.1	64.5	67.7	66.9	64.0	58.3	51.8	48.2		
		MCDB	47.1	47.5	51.1	59.9	65.4	69.8	73.7	72.3	67.7	60.5	53.5	49.5		
	10%	WB	44.3	44.3	46.8	52.6	58.0	62.4	66.0	65.3	62.3	56.6	50.0	46.1		
		MCDB	45.4	45.9	48.9	56.8	62.9	66.5	71.2	70.1	65.7	58.6	51.4	47.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	6.9	8.4	10.1	9.6	9.4	9.5	10.1	9.3	8.4	6.6	6.1		
		MCDBR	6.5	8.9	12.2	17.1	15.7	15.5	15.4	15.8	14.0	10.4	7.5	6.2		
	5% WB	MCWBR	6.2	7.3	8.3	10.1	9.2	9.2	8.5	8.4	8.3	7.2	6.4	6.0		
		MCWBR	6.6	8.2	11.2	15.1	13.5	14.2	13.9	13.3	12.1	10.2	7.4	6.3		
(40) Clear-Sky Solar Irradiance	taub	taub	0.330	0.348	0.376	0.405	0.403	0.401	0.412	0.406	0.388	0.363	0.342	0.322		
		taud	2.312	2.336	2.281	2.251	2.307	2.331	2.312	2.344	2.383	2.406	2.346	2.306		
	Ebn at Noon	Ebn at Noon	199	236	253	261	268	270	264	258	248	228	193	177		
		Edh at Noon	19	26	33	38	38	38	38	35	30	24	19	16		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	186	390	794	1319	1640	1716	1633	1359	954	528	226	141		
	RadStd	RadStd	18	62	80	127	125	120	180	106	91	52	21	16		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page